System and Data Flows

Surette Nexus · Surette Real Estate, Waterville, Maine

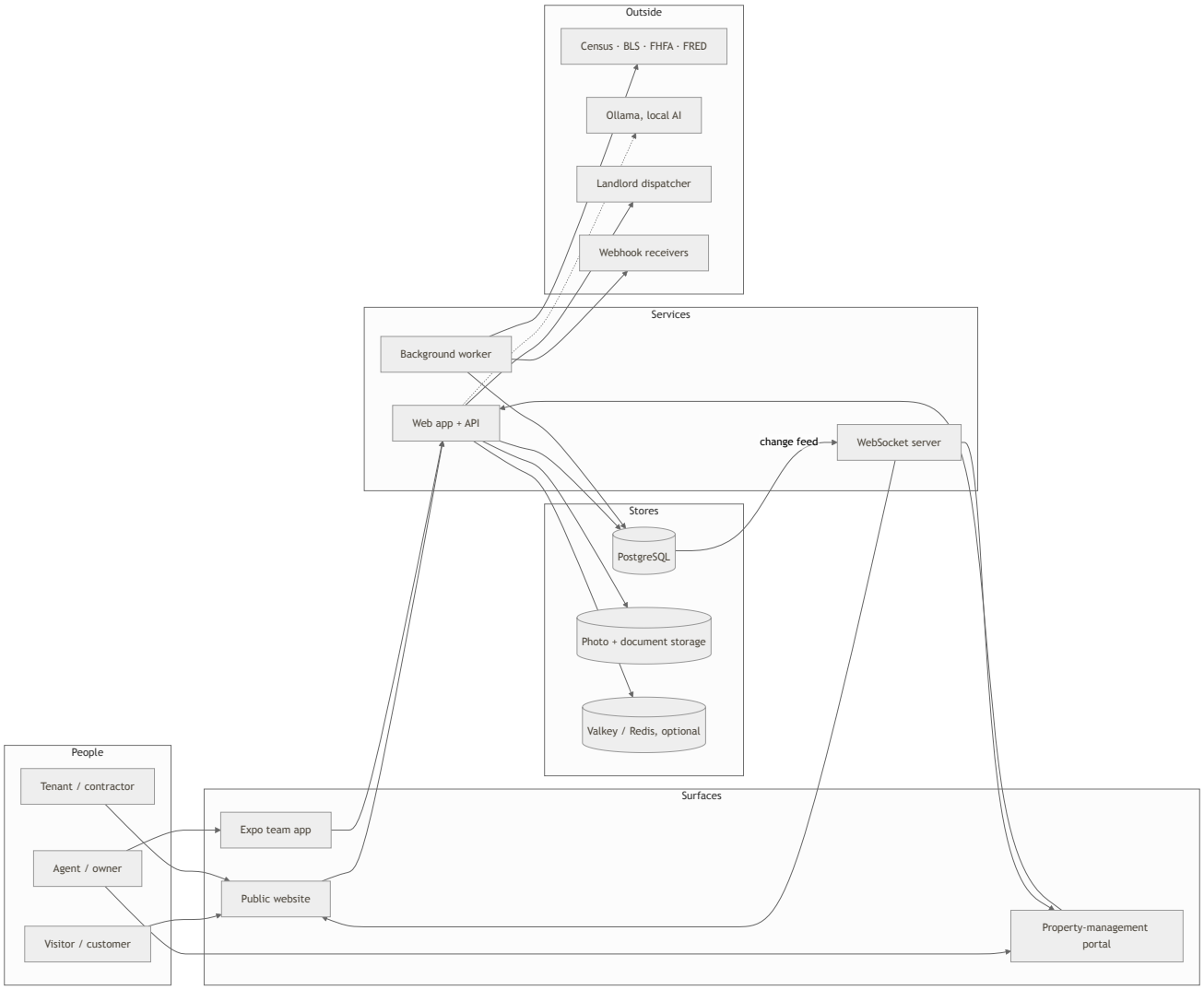
Generated October 18, 2026 from docs/system-flows.md

1. The big picture
2. A visitor browses the site
3. Signing in (two separate entrances)
4. A visitor becomes a lead
5. The AI assistant (chatbot)
6. Listing and photo upload (portal or team app)
7. A deal from contract to closing
8. Showing requests
9. Website analytics (first-party, consent-based)
10. Market and government data
11. Events, queue and webhooks
12. Live updates
13. Maintenance requests and contractors
14. Customer privacy controls
15. When things fail

How each part of Surette Nexus works step by step: who starts it, which services and tables it touches, what is stored, who can see it, and what happens when something fails. For the infrastructure behind these flows (gateway, queue, caching, scaling, monitoring), see system-design.md.

Status as of 2026-10-16.

1. The big picture



Data inventory

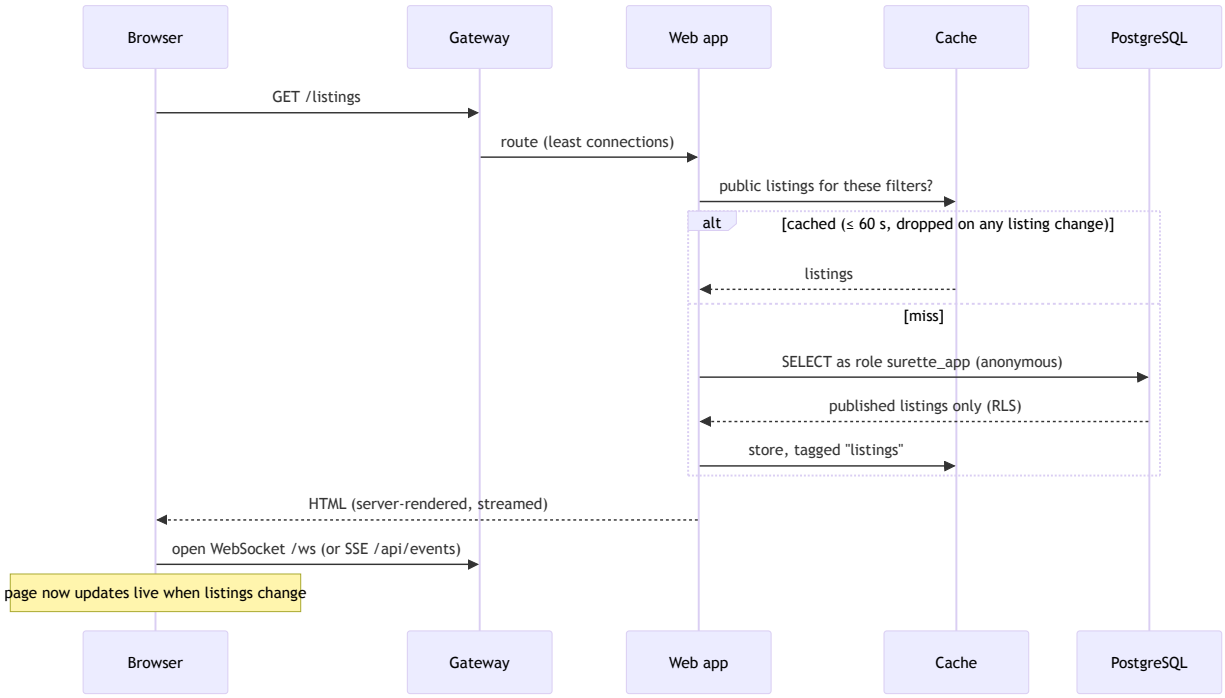
Data	Table(s)	Who can see it	Kept for
Listings and photos	`properties`, `media_assets`, storage `listings/`	Everyone (published), staff (drafts)	Until deleted by staff
Customer accounts	`users` (role `client`)	The customer; their assigned agent; owner	Until the customer deletes it
Staff accounts	`users` (role `agent` / `admin`)	Owner	Until deactivated
Saved homes, searches, comparisons, Dual-Move plans	`collections*`, `saved_searches`, `search_history`, `property_comparisons`, `dual_move_scenarios`	The customer (and their agent for shared collections)	Until the customer deletes them or the account

Data	Table(s)	Who can see it	Kept for
Leads (inquiries)	`leads`, `lead_notes`, `consent_records`	Assigned agent, owner	Until staff delete them
Deals	`transactions`, `transaction_milestones`, `transaction_tasks`	The deal's agent, owner; the client sees client-visible milestones	Until deleted by the owner
Deal documents	`transaction_documents`, storage `deals/`	The deal's agent, owner	Until deleted
Email drafts	`communication_drafts`	The deal's agent, owner	Until discarded
Website analytics	`analytics_events`	Owner (full), agents (their listings' engagement)	Project retention, default 395 days
Market & government data	`market_series`, `market_observations`, `gov_dataset_imports`	Everyone	Replaced on refresh
Audit trail	`audit_log` (append-only)	Owner	Kept
Events & jobs	`outbox_events`, `jobs`	System only	7 days after completion
Idempotency keys, rate-limit counters	`idempotency_keys`, `rate_limit_counters`	System only	24 hours / 2 hours

Every request runs as a restricted database role with the caller's identity attached, so the "who can see it" column is enforced by PostgreSQL row-level security, not only by code.

---

2. A visitor browses the site

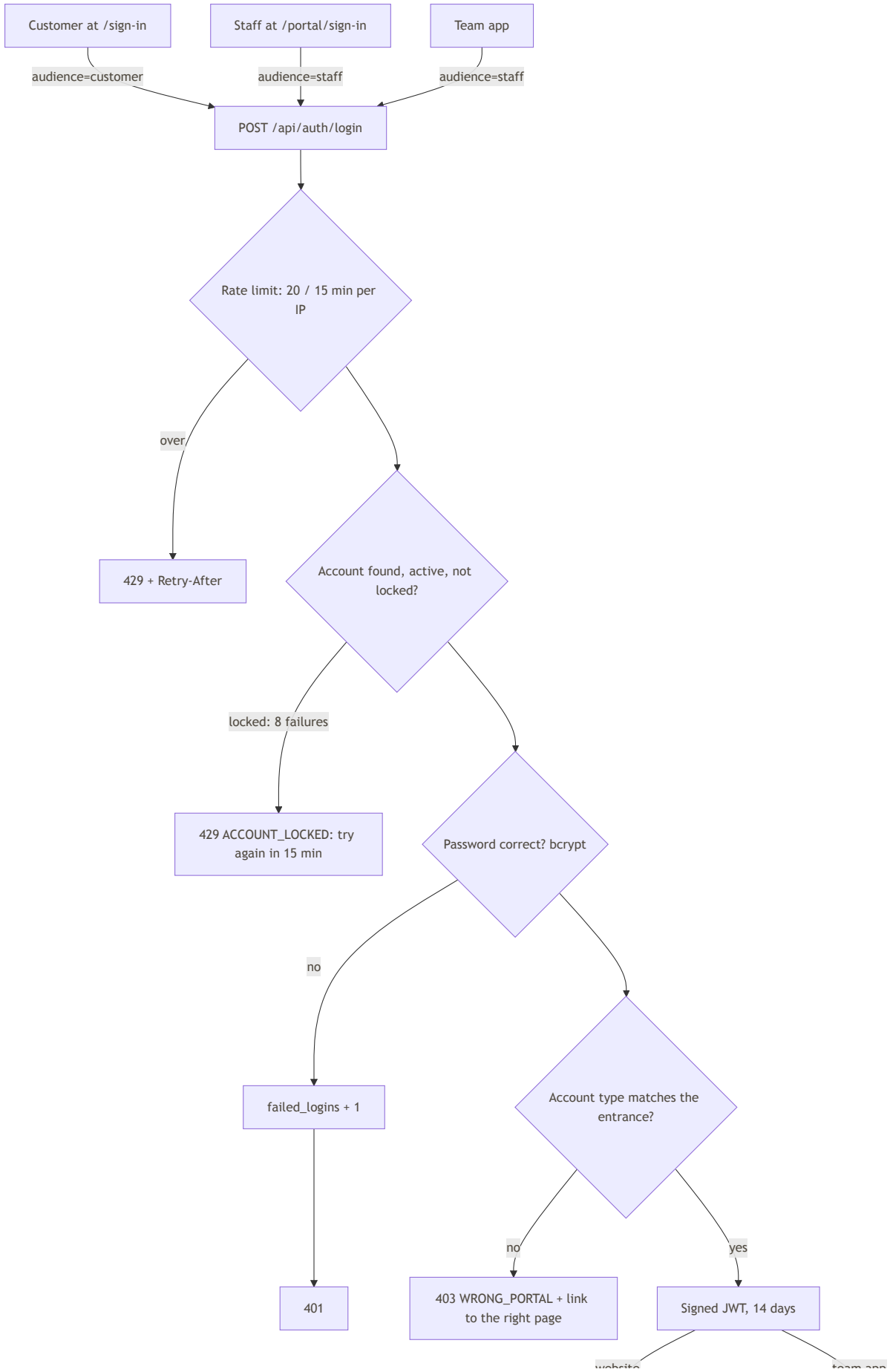


1. Pages render on the server. Public data comes from the in-app cache when possible.
2. The browser opens one live connection per tab. When a listing or photo changes, the browser receives only the id and re-fetches through the API.
3. Images are served as pre-sized WebP renditions (640, 1200 and 2400 px) with a long cache lifetime.
4. The analytics tracker loads but sends nothing until the visitor chooses "Allow" (see §9).

If something is down: cache miss → database. Database down → error page, and readiness takes the instance out of the load balancer. WebSocket server down → the browser falls back to SSE.

---

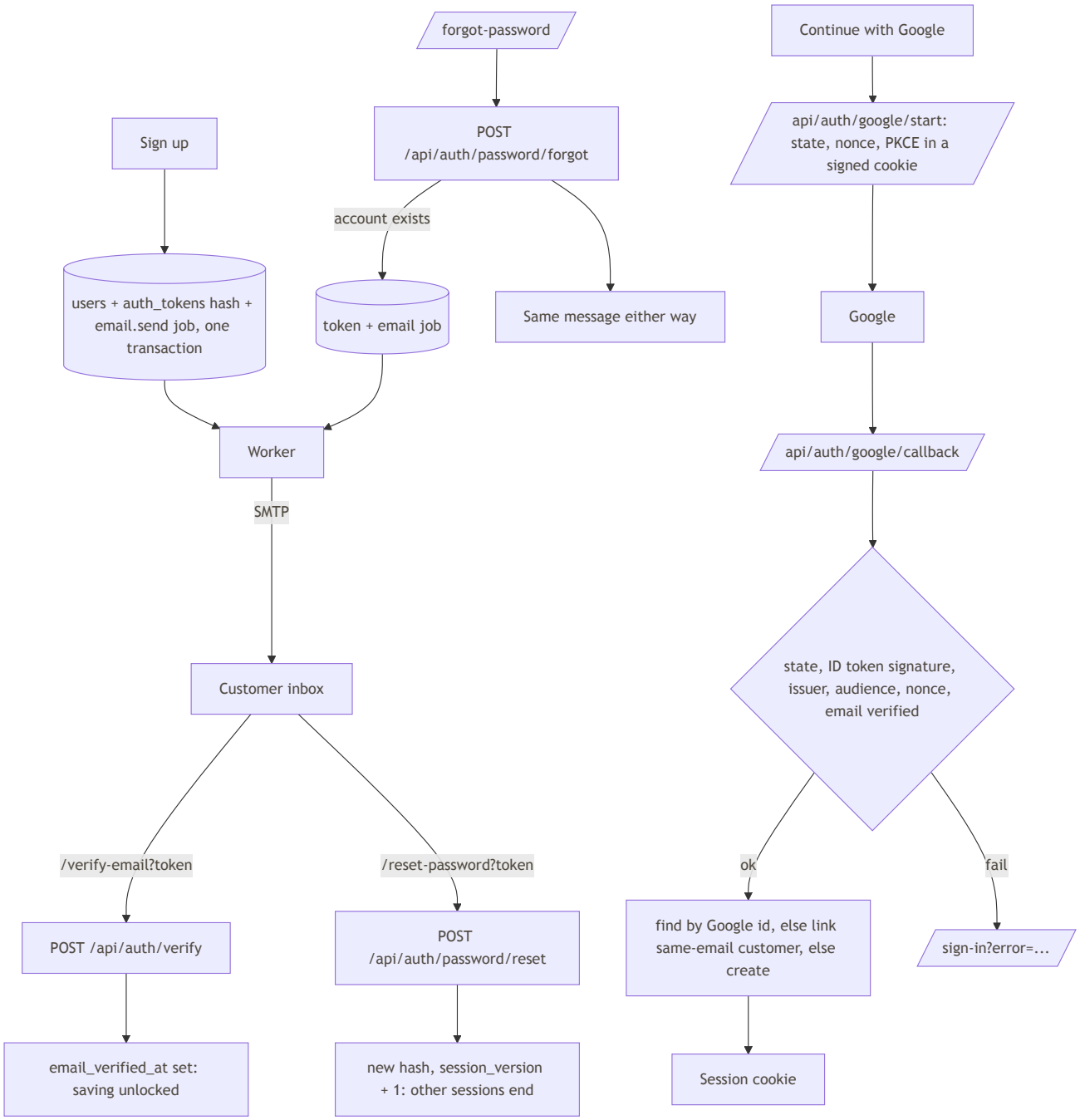
3. Signing in (two separate entrances)


Customer at /sign-in
Staff at /portal/sign-in
Team app
audience=customer
audience=staff
audience=staff
POST /api/auth/login
Rate limit: 20 / 15 min per IP
over
429 + Retry-After
Account found, active, not locked?
locked: 8 failures
429 ACCOUNT_LOCKED: try again in 15 min
Password correct? bcrypt
no
failed_logins + 1
401
Account type matches the entrance?
no
403 WRONG_PORTAL + link to the right page
yes
Signed JWT, 14 days



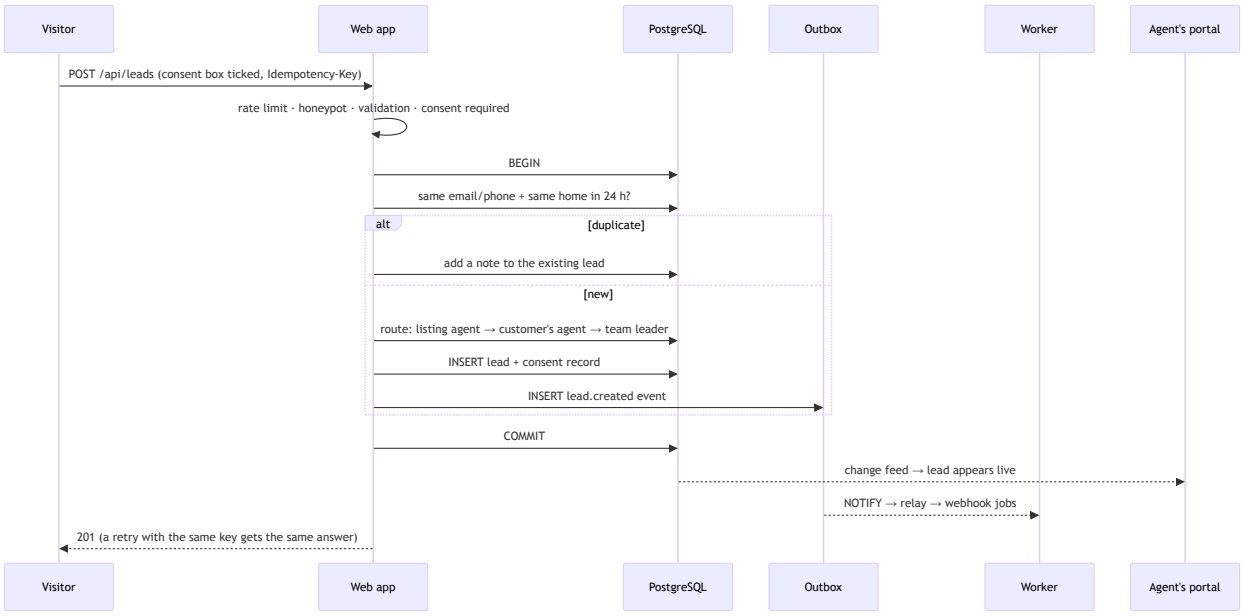
- Every API request re-reads the user row, so deactivation or a role change takes effect immediately.
- Cookie-authenticated writes from another website are refused (CSRF protection in `proxy.ts`).
- Staff accounts are created only by the owner (Portal → Team) or `npm run admin:create`.

Customer email confirmation, password reset and Google



4. A visitor becomes a lead

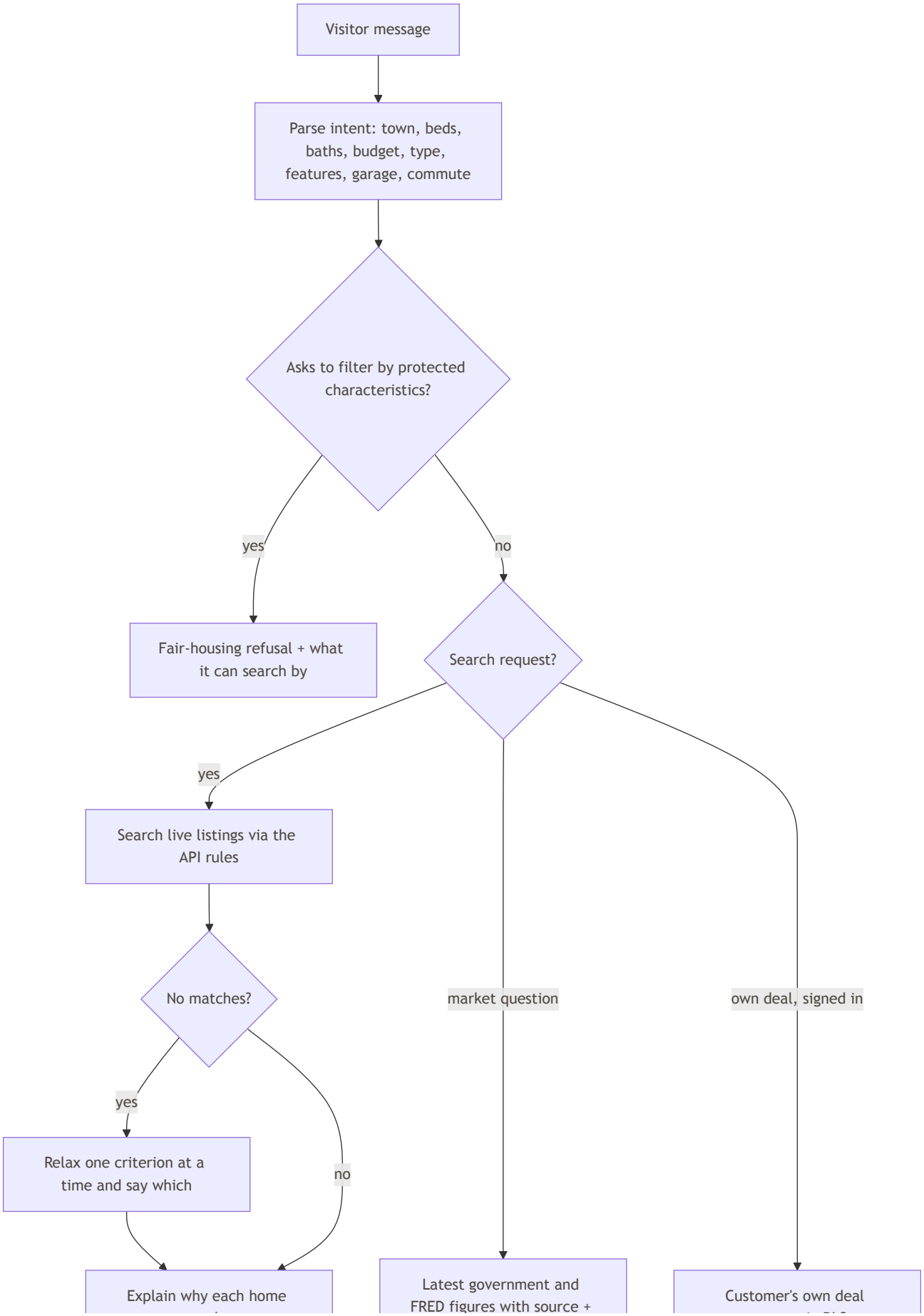
Leads come from the contact page, showing requests, the AI assistant hand-off and Dual-Move "send to an agent".

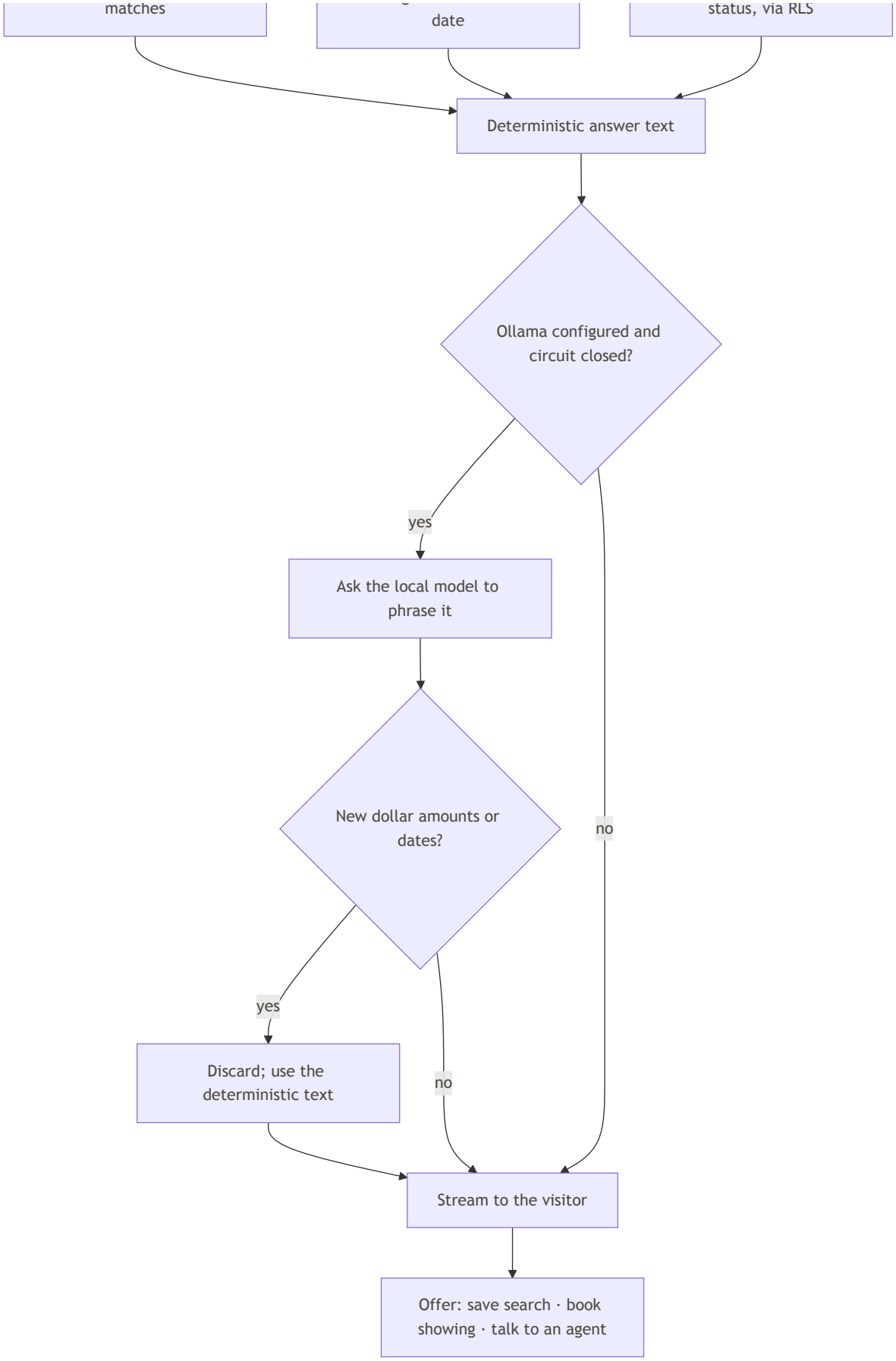


What's stored: name, email and/or phone, message, explicit preferences, the exact consent text and time. Never stored: anything inferred about the person.

---

5. The AI assistant (chatbot)


Visitor message
Parse intent: town, beds, baths, budget, type, features, garage, commute
Asks to filter by protected characteristics?
yes
no
Fair-housing refusal + what it can search by
Search request?
yes
Search live listings via the API rules
No matches?
yes
Relax one criterion at a time and say which
no
Explain why each home
market question
Latest government and FRED figures with source +
own deal, signed in
Customer's own deal

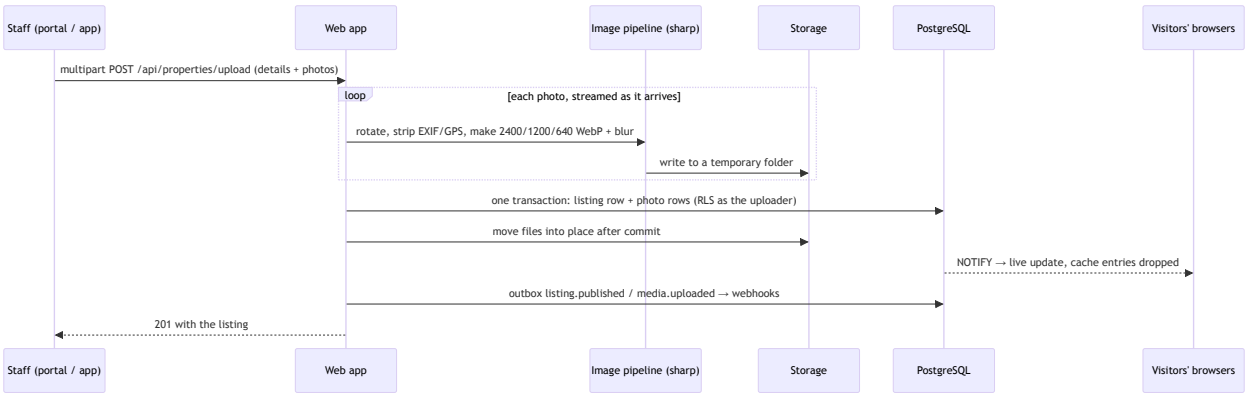

matches
date
status, via RLS
Deterministic answer text
Ollama configured and circuit closed?
yes
no
Ask the local model to phrase it
New dollar amounts or dates?
yes
no
Discard; use the deterministic text
Stream to the visitor
Offer: save search · book showing · talk to an agent

The AI assistant only mentions homes the database returned. AI is optional, runs on your own server and

never sees anything the deterministic answer didn't already contain.

---

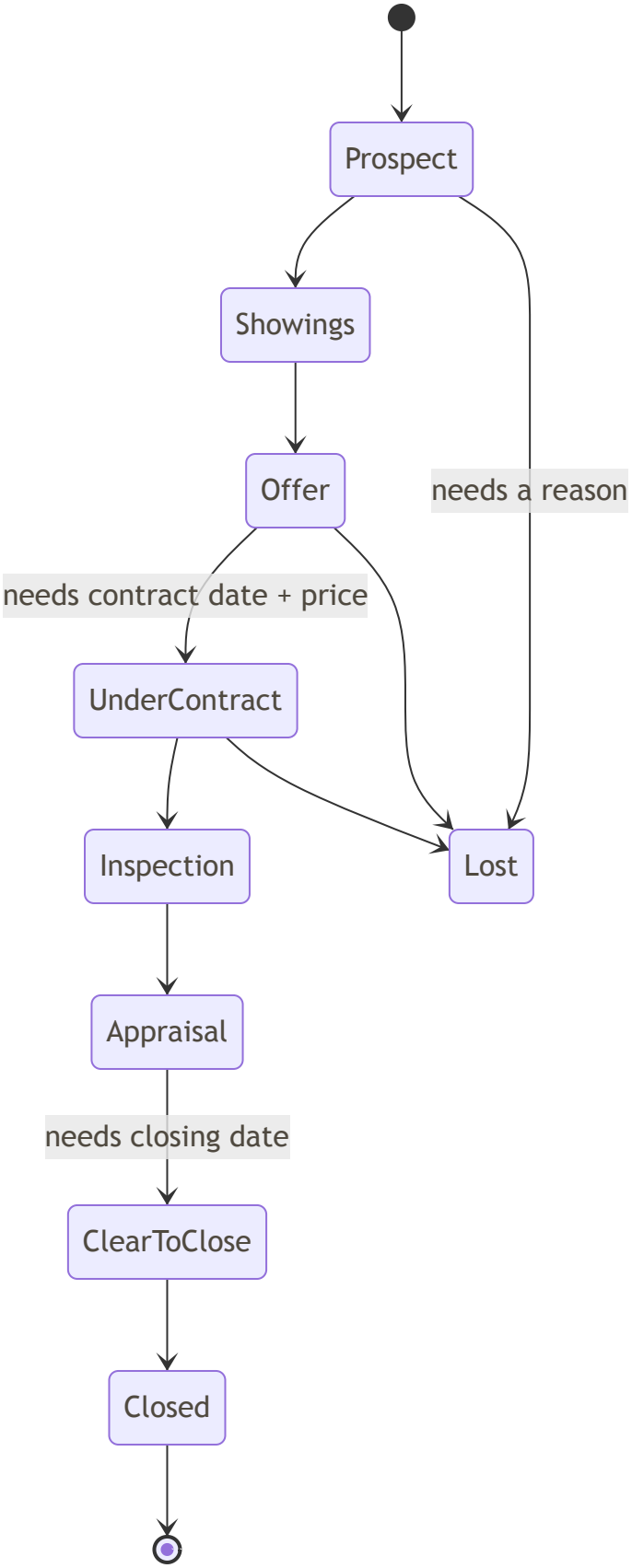
6. Listing and photo upload (portal or team app)



Limits: 40 photos, 30 MB each; JPEG, PNG, WebP, AVIF or HEIC only. Every file is decoded by the image pipeline, so anything that isn't really an image is rejected.

---

7. A deal from contract to closing



1. **Create** (board or from a lead). The side's checklist template is applied: each milestone and task gets a rule such as `contract+10:roll` (10 days after the contract, moved off weekends).
2. **Move stages.** Stage gates block a move until the required dates or price exist.
3. **Dates change.** Rule-based items are recalculated; items someone rescheduled by hand stay put.
4. **Documents.**
    1. Upload a PDF or image; the type is checked from the content.
    2. Text is scanned for price, deposit, dates and inspection days, each with the page it came from.
    3. A person verifies or rejects each value before anything is applied to the deal.
5. **Emails.**
    1. Drafts are generated from the deal's data.
    2. A person must approve the text before it can be marked "sent".
    3. Staff send it from their own email; the system never emails clients by itself.
6. **Record and notify.**
    - Every change is written to the append-only audit log.
    - Stage changes emit `deal.stage_changed`, delivered by webhook.
    - The client sees client-visible milestones in their account.

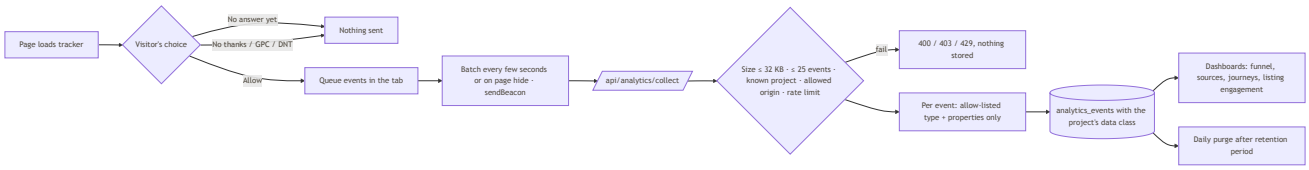
---

8. Showing requests

1. A visitor picks a slot on a listing (open slots come from `/api/showings/slots`).
2. The request is stored as `requested`. A database exclusion constraint makes double-booking a confirmed slot impossible.
3. The request also creates or updates a lead (§4), with consent.
4. The listing agent confirms or declines in the portal. The customer sees the status in their account, and the change is pushed live.

---

9. Website analytics (first-party, consent-based)

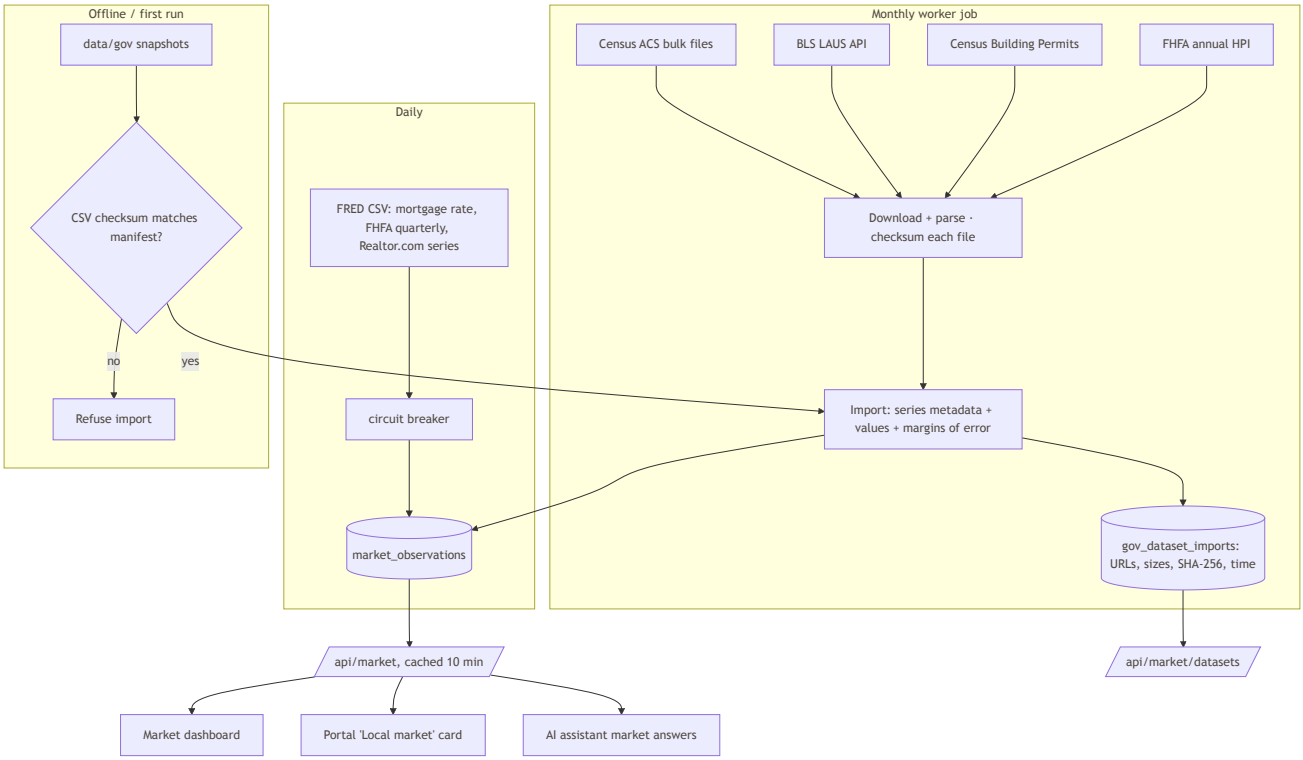


- **Projects:** `nexus-site` (production, this website), `demo-site` (synthetic), `e2e` (tests). The data class comes from the project, never from the browser.
- **Never collected:** form contents, names, emails, phone numbers, full IPs, cookies, or cross-site ids.
- **Identifiers:** a per-tab session id always; a pseudonymous visitor id only after "Allow".

- **Duplicates:** resent batches are ignored by event id.

---

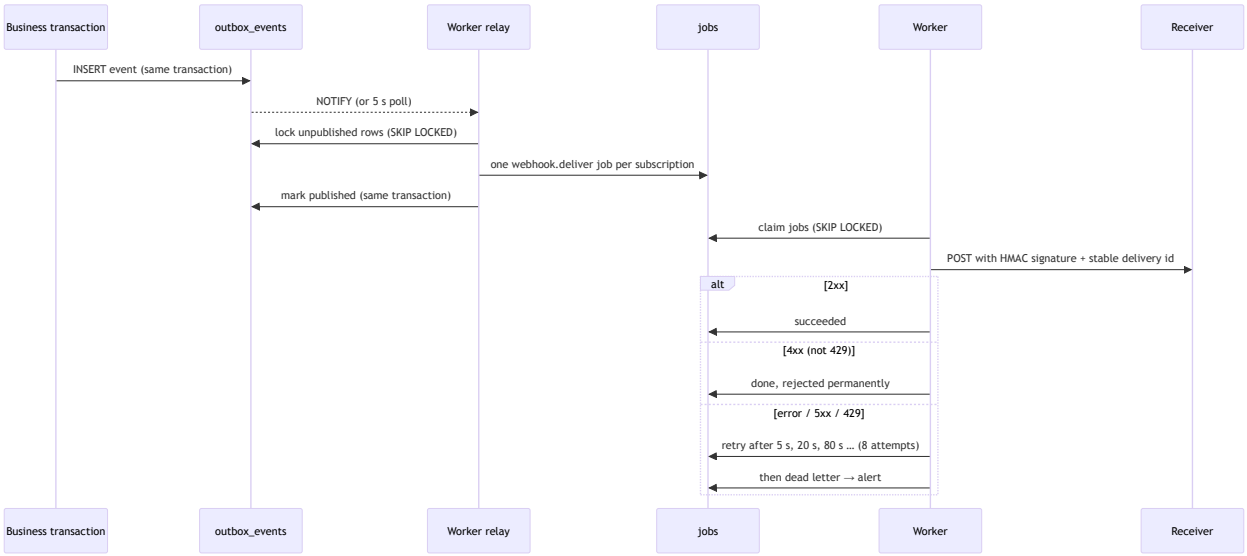
10. Market and government data



- **Downloads fail:** previous values stay, and the series records the error. Nothing is estimated.
- **Gaps stay gaps:** for example, BLS never collected October 2025.
- **Census figures:** shown with their margin of error.

---

11. Events, queue and webhooks



Recurring jobs (scheduled by whichever worker holds the leader lock):

Job	Every	Does
`maintenance.cleanup`	hour	Clears expired rate-limit counters and idempotency keys; removes old finished jobs and published events
`analytics.purge`	day	Deletes analytics events past each project's retention
`market.refresh`	day	FRED series
`govdata.refresh`	30 days	Census, BLS, FHFA downloads

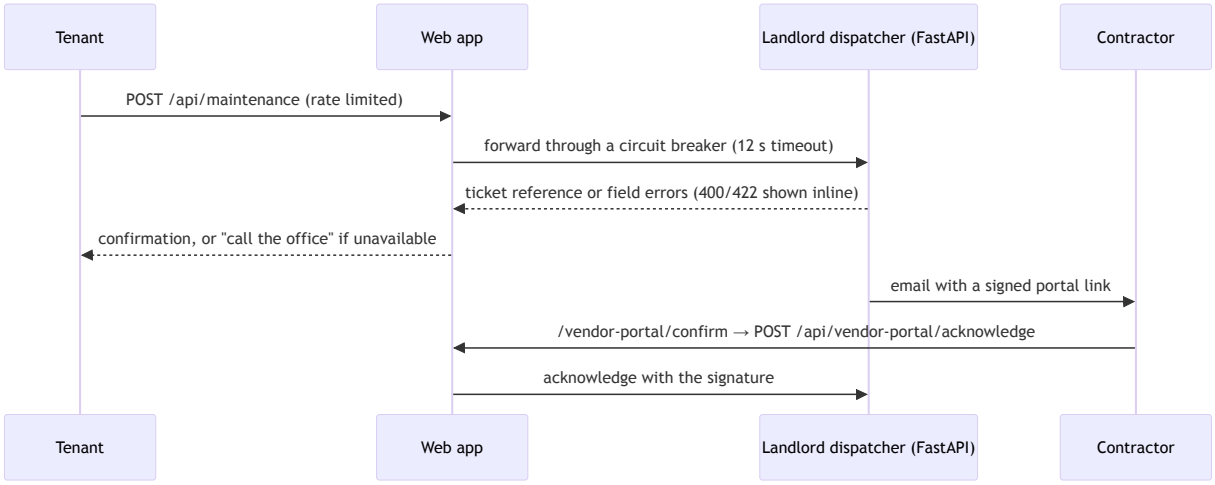
---

12. Live updates

1. A database trigger publishes `{table, op, ids, visibility, team}` on the `surette_events` channel after every relevant change, whoever made it.
2. Each web and realtime instance keeps one LISTEN connection.
3. For each connected client, the shared visibility rules decide whether to forward the event: public listing changes go to everyone; staff changes go to that team's staff; a customer gets their own showings and deal; collection activity goes to its owner and agent.
4. The client re-fetches the changed item through the API, where row-level security applies again.

---

13. Maintenance requests and contractors



Staff see vendor insurance status in Portal → Vendors (read from the dispatcher).

---

14. Customer privacy controls

Action	Where	What happens
Change analytics choice	`/privacy`, banner	Stored in the browser; "No thanks" also removes the visitor id
Export my data	Account → Export	JSON of the account, collections and comments, saved searches, search history, comparisons, Dual-Move scenarios, showings and consent records
Delete my account	Account → Delete (password required)	Account and everything stored for it are deleted. Leads already sent to the team are unlinked; the office can erase those on request
Withdraw contact consent	Ask the office	Consent record updated; staff stop contact

---

15. When things fail

Failure	What visitors see	What staff see	Recovery
One web instance crashes	Nothing (others serve)	—	Restarted by Docker / Kubernetes
Database unreachable	Error page	Portal unavailable; critical alert	Failover (CloudNativePG) or restart

Failure	What visitors see	What staff see	Recovery
Worker down	Nothing	Webhooks and refreshes wait; alert after 5 min	Start a worker; queued work drains
Ollama down	Built-in answers	—	Breaker retries every 30 s
Census / BLS / FHFA / FRED down	Last stored data with its date	Series shows its last error	Next scheduled refresh
Dispatcher down	"Call the office" message	Vendors page unavailable	Breaker retries every 30 s
Redis down	Nothing	Warning alert	Falls back to Postgres limits and in-app cache
WebSocket server down	Updates continue over SSE	—	Restart
Traffic spike	Slightly slower pages	Latency and pool alerts	Autoscaling adds web pods